

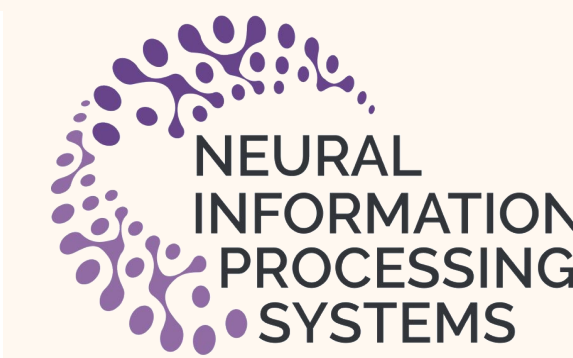


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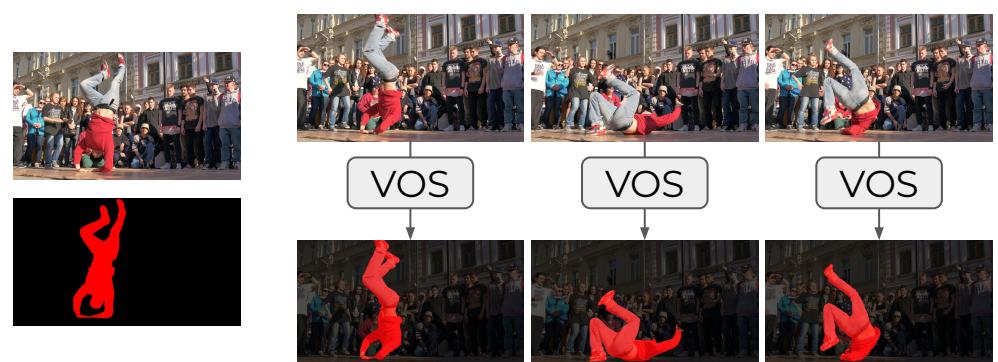
Test-Time Training for Matching-based Video Object Segmentation

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* Equal contribution



Motivation



Video Object Segmentation (VOS) is a dense prediction task over time and space $\rightarrow$ collecting training data is highly demanding

We improve the generalization under **distribution shifts** by fine-tuning the model at **test time** through a single test video

Contributions and highlights

A novel **test-time training** strategy:

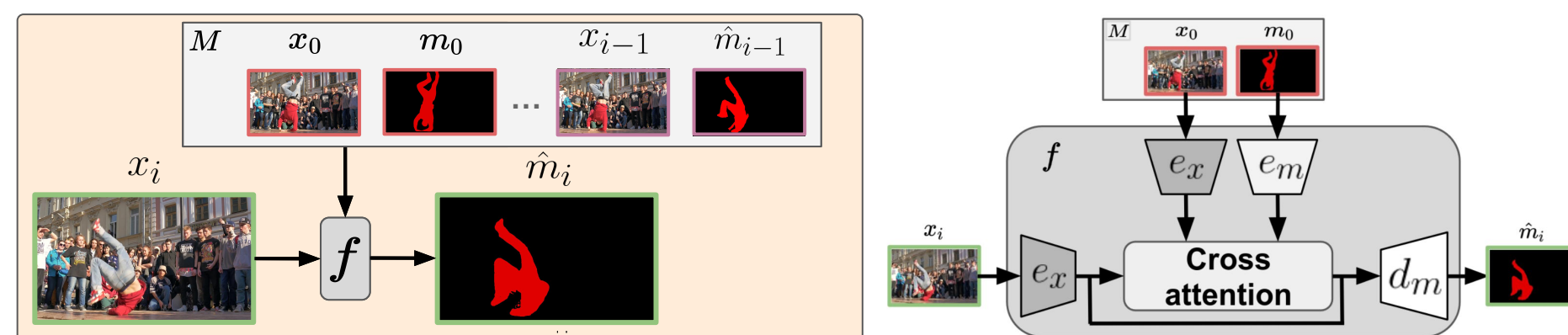
1. tailored for matching-based video object segmentation
2. using only the ground-truth mask for the first frame
3. exploring the time dimension, with **mask cycle consistency**

Boosting performance under two type of extreme distribution shifts

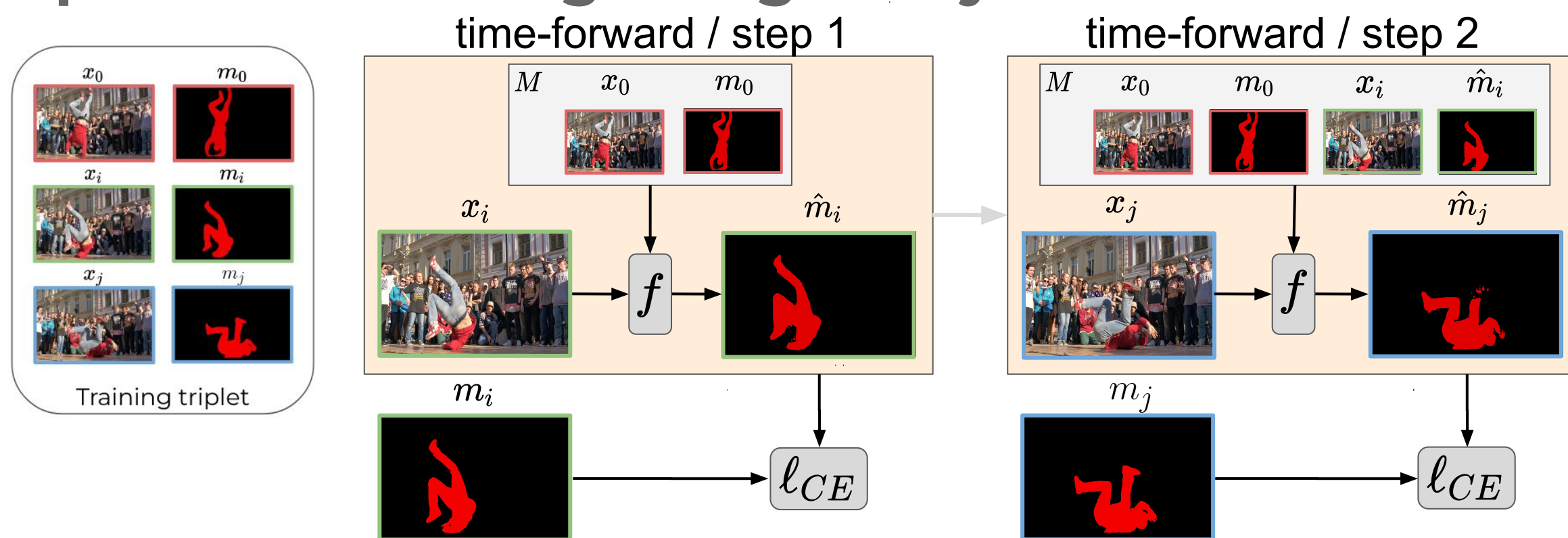
DAVIS-C dataset: 14 corruptions and style changes - 3 levels of strength

Matching-based methods

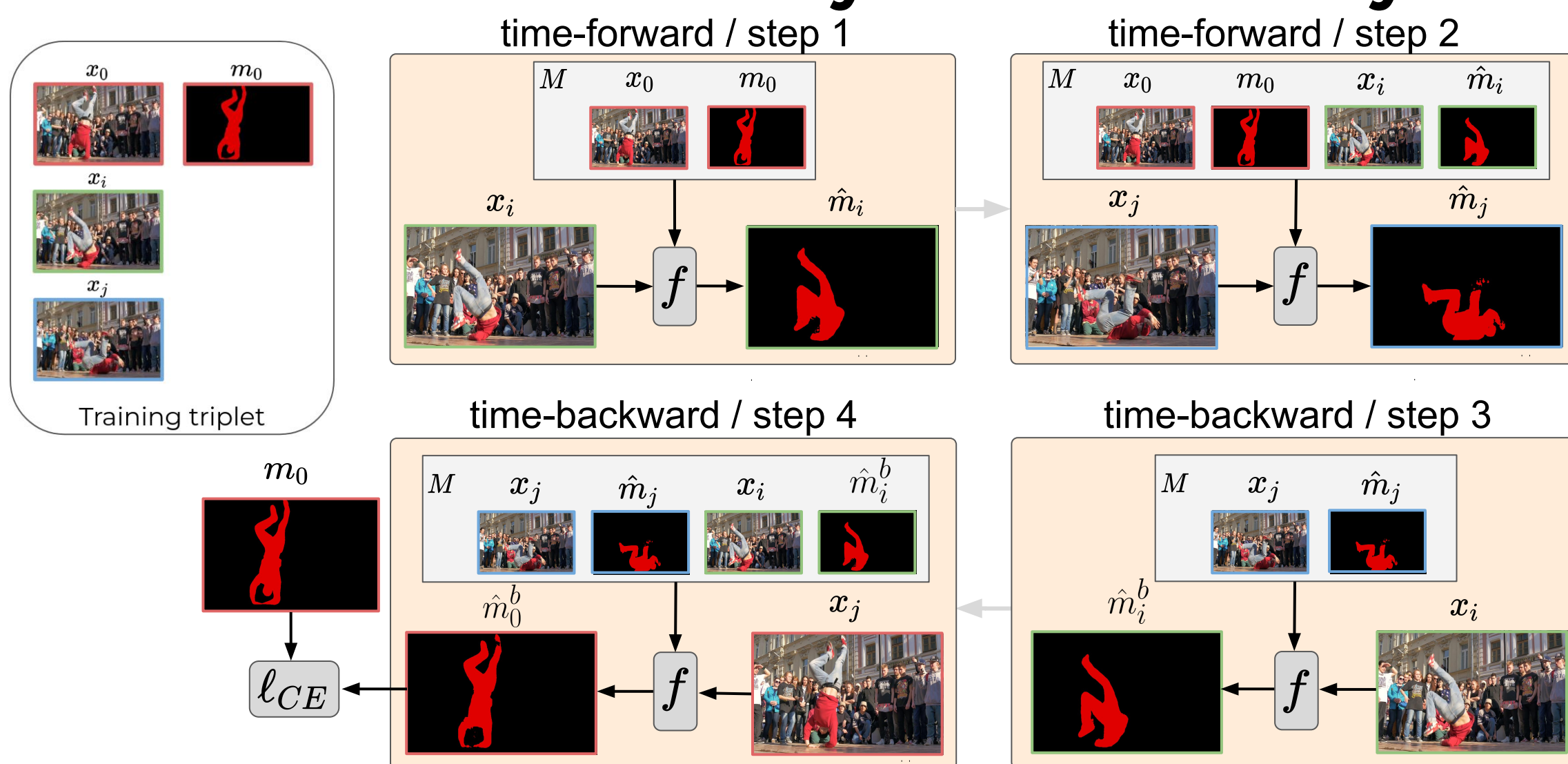
Inference



Supervised training using many annotated frames



Test-Time Mask Cycle Consistency



Sim-to-real transfer



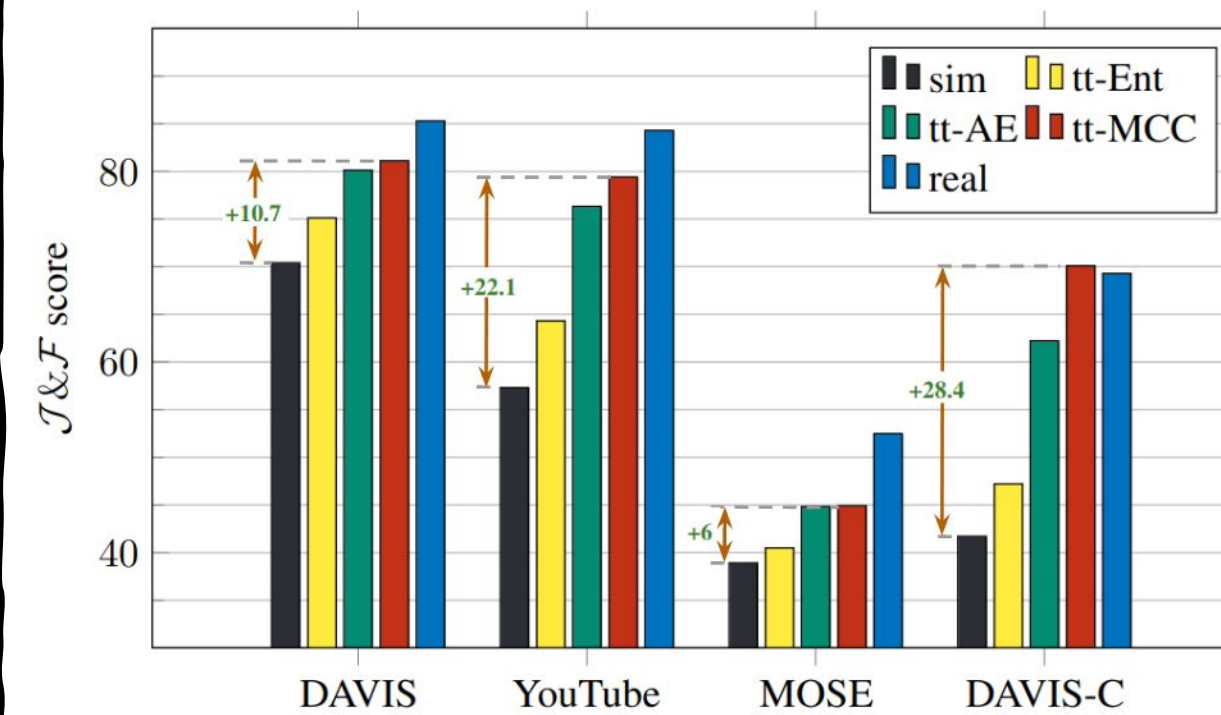
Frames and ground-truth masks



STCN



STCN + tt-MCC (ours)



Corruption - Stylisation



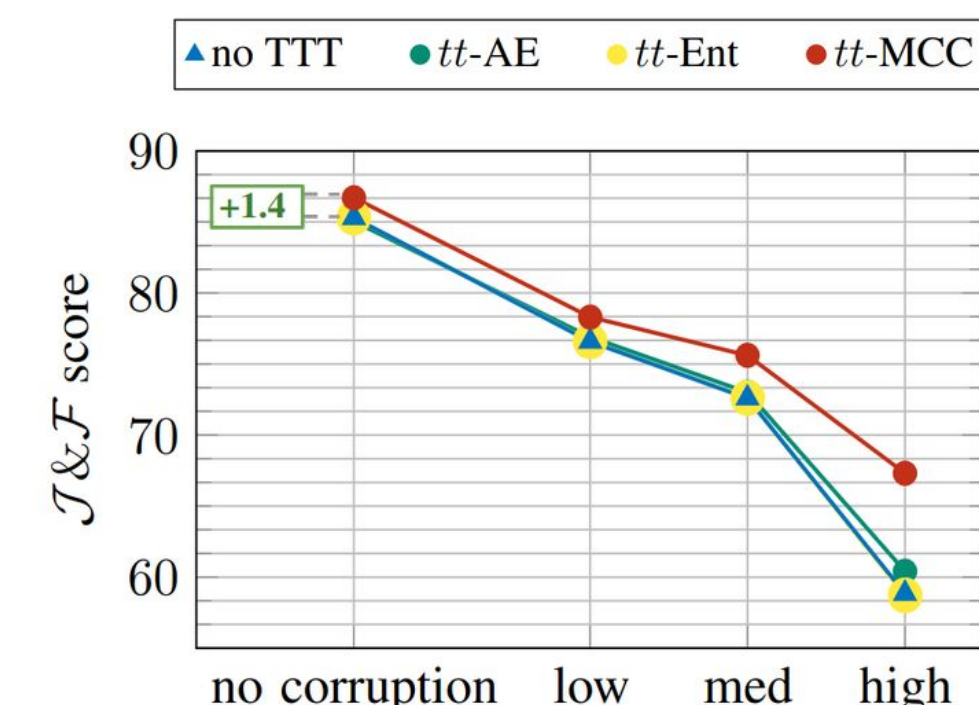
Frames and ground-truth masks



STCN



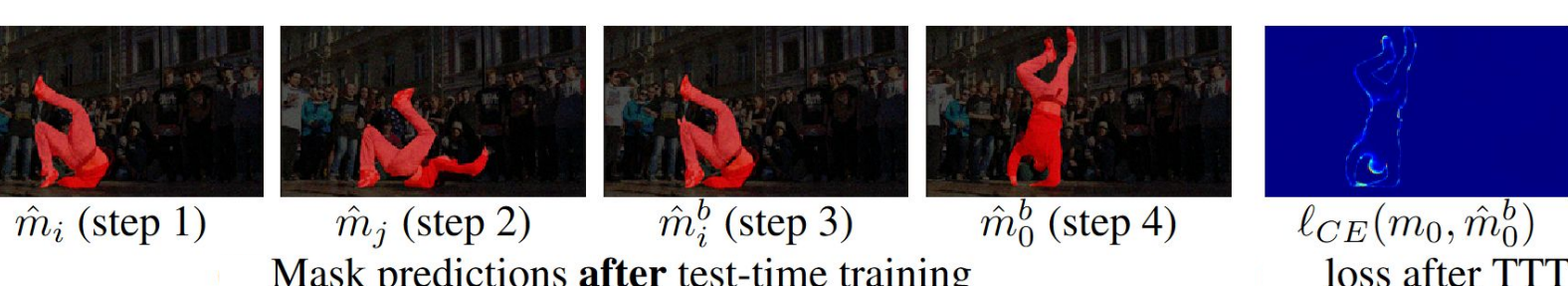
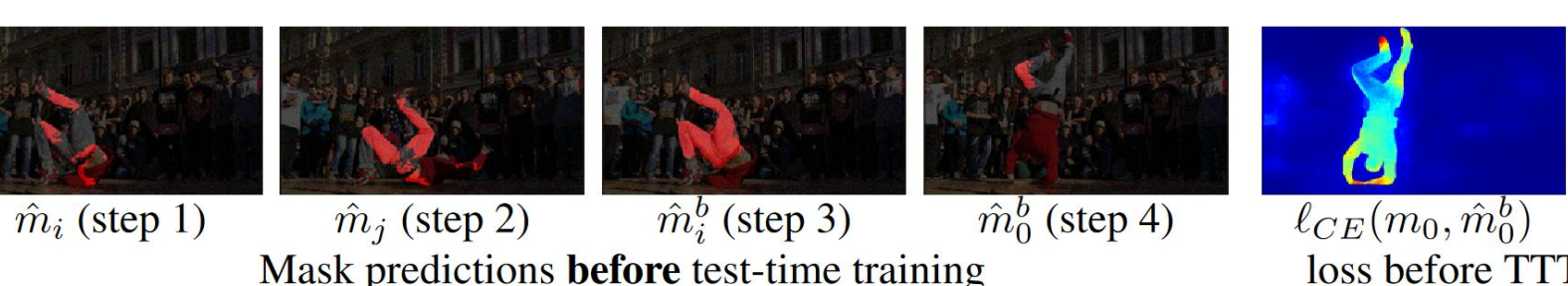
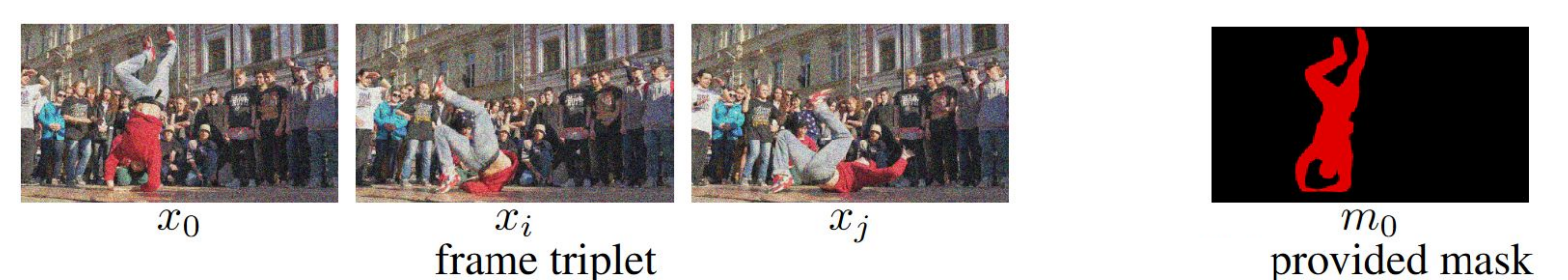
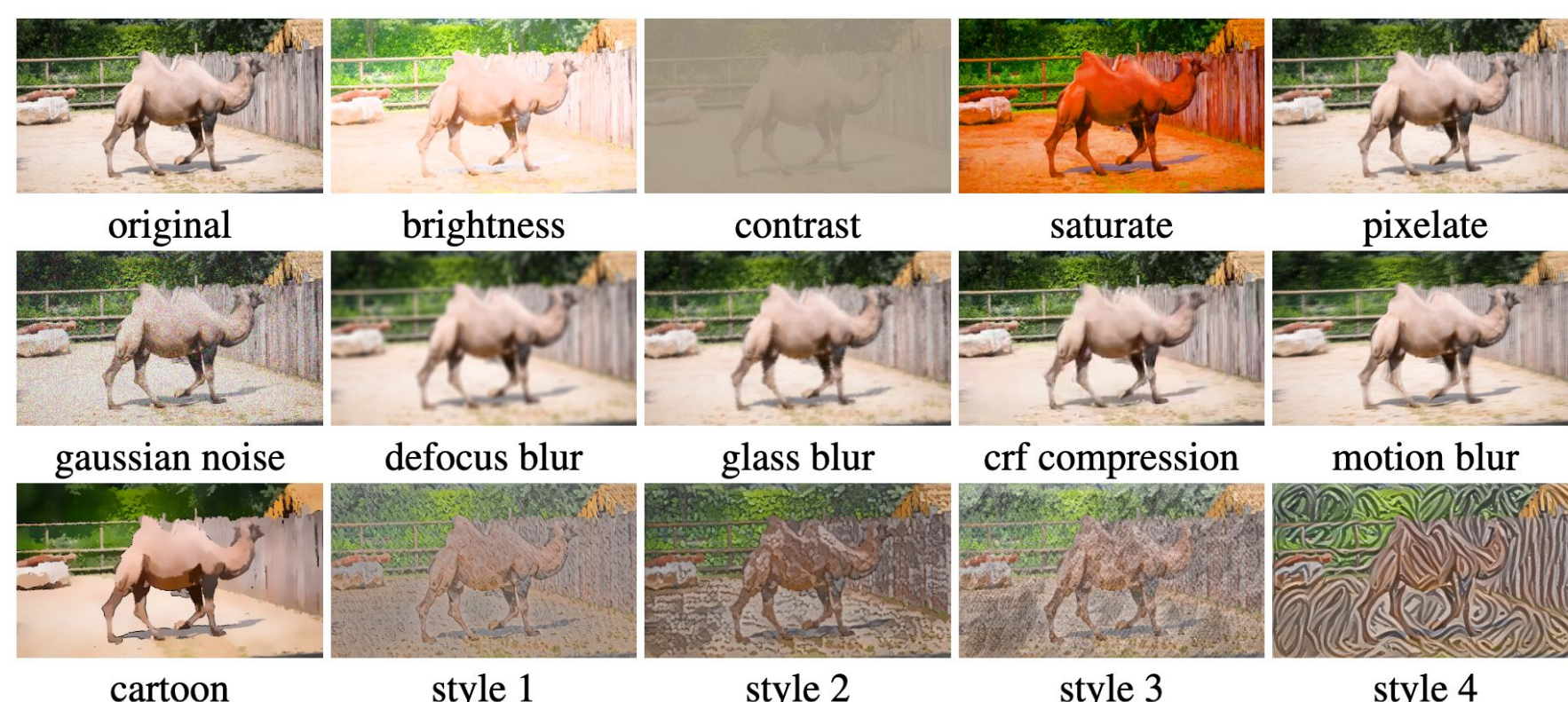
STCN + tt-MCC (ours)



SoTA Comparison

Method	Training without real videos				Corrupted test examples (DAVIS-C)				
	DAVIS	DAVIS-C	YouTube-VOS	MOSE	no corr.	low	med	high	avg
HODOR [1]	77.5 [†]	—	71.7 [†]	—	81.3 [†]	69.0	64.5	55.2	65.0
STCN [9]	70.4	41.7	57.3	38.9	85.3	76.6	72.6	58.8	73.3
STCN + tt-MCC (ours)	81.1	70.1	79.4	44.9	86.7	78.3	75.6	67.3	77.0
XMem [6]	78.1	53.9	65.6	40.9	87.7	80.4	77.3	69.4	78.7
XMem + tt-MCC (ours)	82.1	70.1	78.9	44.7	88.1	81.7	78.9	72.2	80.2

DAVIS-C dataset



Further Analysis

